

Is the sequence editor sensitive to learning rate?

Separating optimization effects from random variation

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The question is about fair comparisons

- ▶ The learning rate controls how far the model moves after each training signal.
- ▶ Too small a rate can make a healthy model look weak simply because training ends too soon.
- ▶ Random seeds cause a different problem: initialization and example order vary between otherwise identical runs.
- ▶ We need to measure both before attributing a score change to a new architecture or objective.

Why it matters: width, counterfactual losses, LDAD, GAR, and hierarchy may each prefer different update sizes.

Only the peak learning rate changes

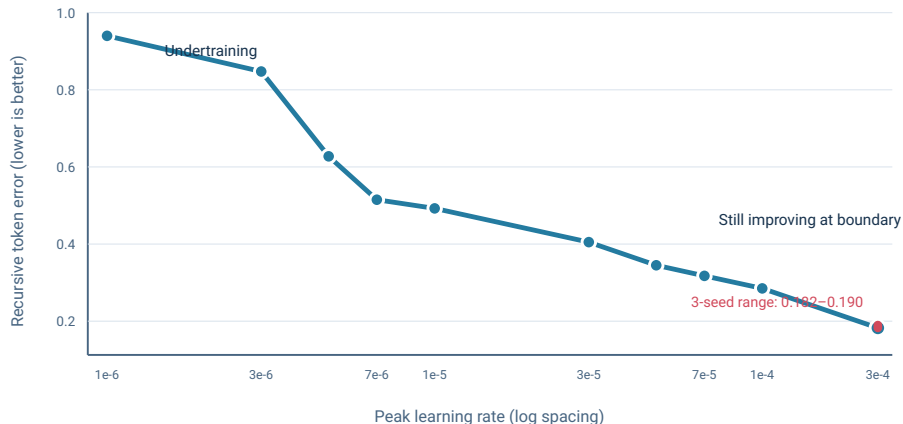
- ▶ Every cell uses fixed mixed corruptions, token-aligned EMA targets, $K=0$, d256, batch eight, and no dropout.
- ▶ Training contains 6,000 unique iGSM problems over three epochs, or 18,000 presentations.
- ▶ The sweep spans 10^{-6} through 3×10^{-4} under the same cosine schedule.
- ▶ Nine new seed-zero cells are combined with the existing three-seed endpoint at 3×10^{-4} .
- ▶ Each completed checkpoint is evaluated on 256 held-out mixed-edit examples.

What to notice: this isolates optimization sensitivity for the current primitive, not for every future method.

Performance improves smoothly across the requested range

Larger updates keep improving the selected edit model

Same recipe, data, schedule, and seed except for the three-seed endpoint



Learning-rate effects dwarf seed noise

Condition	Matched error	Recursive error	Action ratio
10^{-6}	.916	.941	1.04
10^{-5}	.411	.492	1.76
10^{-4}	.169	.285	3.31
3×10^{-4} , seed 0	.095	.182	5.22
3×10^{-4} , seeds 1/2	.100/.096	.190/.185	5.02/5.22

- ▶ Moving from 10^{-4} to 3×10^{-4} cuts recursive error by about 36%.
- ▶ The full seed range at 3×10^{-4} is only about 4.3%.
- ▶ Gradient norms remain benign, so low rates undertrain rather than explode.

The evidence changes how methods must be compared

- ▶ The large exposure and width gains remain larger than measured seed noise, but width needs its own learning-rate cross-check.
- ▶ $K=1$'s 3–4% recursive penalty is comparable to seed spread and must be retested across rate and loss weight.
- ▶ LDAD and GAR alter loss scale, so a common untuned rate would be especially unfair to them.
- ▶ Identical fixed-versus-fresh corruption results remain strong because those cells share the same parameterization and objective.

Why it matters: systematic optimization sensitivity is a confound, but it is not random fluctuation.

Next, bracket the optimum before adding complexity

- ▶ Reuse 3×10^{-4} and test 2×10^{-4} , 5×10^{-4} , 7×10^{-4} , and 10^{-3} until the curve turns over.
- ▶ Confirm the selected primitive rate with at least three seeds.
- ▶ Give d512 and $K=1$ separate three-point learning-rate controls; cross $K=1$ with weights 0.25 and 1.
- ▶ Tune GAR H1/H4 only after the primitive is fixed, then compare hierarchy against a separately tuned flat control.
- ▶ Attempt MPC, beam search, or CEM only after recursive dynamics and GAR ranking pass their gates.